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FIGURE 1
RESOLUTION MSC.36(63)
(adopted on 20 May 1994)
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.2 When the dynamic stability curve is used, first an auxiliary point A should be determined. For this purpose the amplitude of heeling is plotted to the right along the abscissa axis and a point A' is found (see figure 2). A line AA' is drawn parallel to the abscissa axis equal to the double amplitude of

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SOURCE: MSC.36(63).pdf
point A should be determined. For this purpose the amplitude of heeling is plotted to the right along the abscissa axis and a point A' is found (see figure 2). A line AA' is drawn parallel to the abscissa axis equal to the double amplitude of heeling (AA'= 29 z) and the required auxiliary point A is found. A tangent AC to the dynamic stability curve is drawn. From the point A the line AB is drawn parallel to the abscissa axis and equal to 1 radian (57.3°). From the point B a perpendicular is drawn to intersect with the tangent in point E. The distance BE is equal to the capsizing moment if measured along the ordinate axis of the dynamic stability curve. If, however, the dynamic stability levers are plotted along this axis, BE is then the capsizing lever, and in this case the capsizing moment Mc is determined by multiplication of ordinate BE (in metres) by the corresponding displacement in tonnes
Mc = 9.81 ~ BE (kNm)
. 3 The amplitude of rolling 9 z is determined by means of model and full-scale tests in irregular seas as a maximum amplitude of rolling of 50 oscillations of a craft travelling at 90° to the wave direction in sea state for the worst design

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SOURCE: MSC.36(63).pdf
. 3 The amplitude of rolling 9 z is determined by means of model and full-scale tests in irregular seas as a maximum amplitude of rolling of 50 oscillations of a craft travelling at 90° to the wave direction in sea state for the worst design condition. If such data are lacking the amplitude is assumed to be equal to 15° .
. 4 The effectiveness of the stability curves should be limited to the angle of flooding.
DYNAMIC STABILITY CURVE
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FIGURE 2
1.2 Transient and fo i l - born e m odes
1.2.1 The stability should satisfy the provisions of 2.4 and 2.5 of this Code.
1.2.2.1 The stability in the transient and foil-borne modes should be checked for all cases of loading for the intended service of the craft.
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1.2.2.3
1. 2.3
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The stability in the transient and foil-borne modes may be determined either by calculation or on the basis of data obtained from model experiments and should be verified by full-scale tests by

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The stability in the transient and foil-borne modes may be determined either by calculation or on the basis of data obtained from model experiments and should be verified by full-scale tests by the imposition of a series of known heeling moments by off-centre ballast weights, and recording the heeling angles produced by these moments. When taken in the hull-borne, take-off, steady foil-borne and settling to hull-borne modes, these results will provide an indication of the values of the stability in the various situations of the craft during the transient condition.
The angle of heel in the foil-borne mode caused by the concentration of passengers at one side should not exceed 8°. During the transient mode the angle of heel due to the concentration of passengers on one side should not exceed 12°. The concentration of passengers should be determined by the Administration, having regard to the guidance given at annex 7 to this Code.
One of the possible methods of assessing foil-borne metacentric height (GM) in the design stage for a particular foil configuration is given in figure 3 .
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FIGURE 3

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SOURCE: MSC.36(63).pdf
to the guidance given at annex 7 to this Code.
One of the possible methods of assessing foil-borne metacentric height (GM) in the design stage for a particular foil configuration is given in figure 3 .
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FIGURE 3
GM-nn (2t;~ID-S) +nll (21~~'1H s)
wh_ nIl p.rcenllg. 01 hydloloilload borne bv IranI foil
nH percentege of hvdrofoilload bOlne by aft foil
In .. cf .. rance widlh oHronl foil
lH .. clellance widlh of aft foil
 .. clearance between bOllom of keel and water
 __ Oft ...
..
g .. height of cenll. of graVIty above bottom 01 keel
1 n .. engle at which iranI foil is inchned to horizon'"
til" lingle al which aft foil is Inclined 10 honlon,,1
S . . = height of centre of grav i ty above water
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2 Fully submerged hydrofoils
2.1 Hull-borne mode
2.1.1 The stability in the hull-borne mode should be sufficient to satisfy the provisions of 2.3 and 2.6 of this Code.
2.1.2 Paragraphs 1.1.2 to 1.1.5 of this annex are appropriate to this type of craft in the hull-borne mode.

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SOURCE: MSC.36(63).pdf
2.1.1 The stability in the hull-borne mode should be sufficient to satisfy the provisions of 2.3 and 2.6 of this Code.
2.1.2 Paragraphs 1.1.2 to 1.1.5 of this annex are appropriate to this type of craft in the hull-borne mode.
2.2 Transient mode
2.2.1 The stability should be examined by the use of verified computer simulations to evaluate the craft's motions, behaviour and responses under the normal conditions and limits of operation and under the influence of any malfunction.
2.2.2 The stability conditions resulting from any potential failures in the systems or operational procedures during the transient stage which could prove hazardous to the craft's watertight integrity and stability should be examined.
2.3 Foil-borne mode
The stability of the craft in the foil-borne mode should be in compliance with the provisions of 2.4 of this Code. The provisions of paragraph 2.2

of this annex should also apply.
2.4 Paragraph 1.2.2 of this annex should be applied to this type of craft as appropriate and any computer simulations or design calculations should be verified by full-scale tests.
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appropriate and any computer simulations or design calculations should be verified by full-scale tests.
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STABILITY OF MULTI HULL CRAFT
1 Stability criteria in the intact condition
MSC 63/23/Add.2